NLP Lecture Notes #1.4: Finding Distinctive Terms

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A common and insightful task in text analysis is to compare two collections of documents (corpora) and identify the words that are most characteristic or **distinctive** of each one. For instance, what words do Democrats use more than Republicans? What terms distinguish published essays from unpublished ones? Or scientific papers from news articles?
Finding distinctive terms is useful for:

- **Data Exploration**: It provides a high-level understanding of the linguistic differences between two groups of texts.
- **Feature Selection**: In machine learning, it helps identify the most informative words for a classification task (e.g., distinguishing spam from non-spam email).
- **Understanding Content**: When comparing a corpus A to everything else ($\neg A$), the distinctive terms provide a powerful summary of what corpus A is about.

In this lecture, we will explore several quantitative methods for identifying these distinctive terms. The most intuitive method is to compare the relative frequencies (proportions) of each word across the two corpora.

Definition

For a given word w and a group (corpus) k, we first calculate its frequency as a proportion:

$$f_{w,k} = \frac{\text{count of word } w \text{ in group } k}{\text{count of all words in group } k} = \frac{C(w,k)}{\sum_{w'} C(w',k)}$$

The distinctiveness score for word w is then the simple difference between its frequencies in the two groups (e.g., group 1 and group 2):

$$\text{Score}(w) = f_{w,1} - f_{w,2}$$

A large positive score means the word is distinctive of group 1, while a large negative score means it is distinctive of group 2.

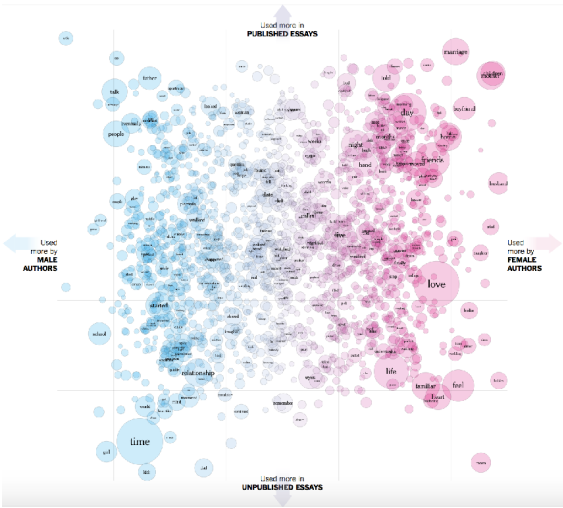

Figure 1: A bubble chart visualizing word usage in published and unpublished essays, with blue bubbles representing male authors and pink bubbles representing female authors, highlighting differences in common themes and vocabulary.

Example

[title=Democrat vs. Republican Speech] Suppose we have two corpora of speeches.

- **Corpus D (Democrat)**: 100,000 total words. The word "women" appears 500 times.
- **Corpus R (Republican)**: 100,000 total words. The word "women" appears 100 times.

The frequencies are:

$$f_{\text{women,D}} = \frac{500}{100,000} = 0.005$$

$$f_{\text{women,R}} = \frac{100}{100,000} = 0.001$$

The difference is $0.005 - 0.001 = 0.004$. This positive score suggests "women" is more distinctive of the Democratic speeches in this sample.

Drawbacks: This method is simple and interpretable, but it has a major drawback: it tends to overemphasize very high-frequency words (like "the", "and") where even small proportional differences can result in a large absolute difference. It also doesn't tell us if the observed difference is statistically meaningful or just due to random chance.

Chi-Square (X^2) Test To address the issue of statistical significance, we can use a statistical test like the **Chi-Square (X^2) test**. This test helps us determine whether there is a significant dependence between two categorical variables. In our case, the variables are:

1. **Word Identity**: Is the word 'robot' or not 'robot'?
2. **Corpus Identity**: Is the document from Corpus 1 (e.g., sci-fi) or Corpus 2 (e.g., not sci-fi)?

The test is based on a **2x2 contingency table** that shows the observed counts.

Example

[title=Contingency Table for "robot"] Let's see if the word "robot" appears significantly more often in a sci-fi corpus than in a general fiction corpus.

	Count of "robot"	**Count of other words**	**Total**
Corpus 1 (Sci-fi)	104 (A)	100,000 (B)	100,104
Corpus 2 (Gen-fi)	2 (C)	150,000 (D)	150,002
Total	106	250,000	250,106

The X^2 formula compares the **Observed** counts (O) in each cell to the **Expected** counts (E) we would see if there were no relationship between the word and the corpus. The formula is:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

A large χ^2 value indicates that the observed counts are very different from what we'd expect by chance, meaning the word's usage is dependent on the corpus (i.e., it is distinctive).

Drawbacks: The X^2 test is very powerful, but it assumes that each observation is independent. This assumption is violated in language due to **burstiness**—the tendency for words to appear in clumps. If a single document in a corpus mentions "Dracula" 500 times, the X^2 test might mistakenly conclude "Dracula" is highly characteristic of the entire corpus, when it's really just characteristic of that one document.

Mann-Whitney Rank Sum Test To account for the burstiness of language, we can use a non-parametric test like the **Mann-Whitney U test** (also known as the rank sum test). This test is more robust to outliers.

Definition

Instead of comparing raw proportions, the Mann-Whitney test compares the **ranks** of observations from two groups. The null hypothesis is that a randomly selected value from group A is equally likely to be greater or less than a randomly selected value from group B.

Application to Text:

1. **Chunking**: Divide both corpora (A and B) into equal-sized chunks of text (e.g., 500 words each).
2. **Counting**: For a specific word (e.g., "Dracula"), count its frequency in every single chunk across both corpora.
3. **Ranking**: Combine all these counts into one large list and rank them from smallest to largest.
4. **Summing Ranks**: Sum the ranks belonging to chunks from Corpus A (R_A) and the ranks belonging to chunks from Corpus B (R_B).
5. **Testing**: The test determines if there is a statistically significant difference between the two sums of ranks.

Example

Handling Burstiness with Mann-Whitney Imagine the word "Dracula" appears 500 times in one chunk of Corpus A, and 0 times in all other chunks of A and B.

- X^2 **Test**: The large total count (500) would lead to a very high χ^2 value, suggesting high distinctiveness.
- **Mann-Whitney Test**: Most chunks have a count of 0, so they will all share a low rank. The one chunk with a count of 500 will get the single highest rank. The sum of ranks for Corpus A will likely not be significantly different from Corpus B, correctly identifying that the word is not broadly distinctive of the corpus, but is just an outlier in one document.

Example

Difference in Proportions You are comparing tweets from two politicians, @AOC and @realDonaldTrump.

- **AOC Corpus**: 50,000 total words. The word "climate" appears 250 times.
- **Trump Corpus**: 80,000 total words. The word "climate" appears 40 times.

Calculate the difference in proportions for the word "climate". Which politician is this word more distinctive of?

First, we calculate the frequency proportion for "climate" in each corpus.

1. **AOC Corpus**:
$$f_{\text{climate,AOC}} = \frac{250}{50,000} = 0.005$$
2. **Trump Corpus**:
$$f_{\text{climate,Trump}} = \frac{40}{80,000} = 0.0005$$

Next, we calculate the difference:

$$\text{Score(climate)} = f_{\text{climate,AOC}} - f_{\text{climate,Trump}} = 0.005 - 0.0005 = 0.0045$$

Since the score is positive, the word "climate" is more distinctive of the **@AOC corpus**.

Example

Interpreting Statistical Tests You run a Chi-Square test and a Mann-Whitney test to find words that distinguish novels written in the 19th century from those written in the 21st century. For the word "email", you get the following results:

- **Chi-Square Test**: Extremely high χ^2 value, p-value < 0.001.
- **Mann-Whitney Test**: Also a very significant result, p-value < 0.001.

For the name "Bartholomew", you get:

- **Chi-Square Test**: High χ^2 value, p-value < 0.001.
- **Mann-Whitney Test**: Not a significant result, p-value $= 0.35$.

How do you interpret the results for "Bartholomew"? What does the discrepancy between the two tests suggest about how this word appears in your corpus?

The results for "Bartholomew" suggest a classic case of **word burstiness**.

- The significant Chi-Square test indicates that, overall, the total count of "Bartholomew" is significantly higher in one corpus (likely the 19th-century novels) than would be expected by chance.
- The non-significant Mann-Whitney test indicates that the word is not used consistently more often across the documents (chunks) of that corpus.

Interpretation: The most likely scenario is that one or very few 19th-century novels in the corpus feature a main character named Bartholomew, leading to a very high frequency of the name in those specific books (it's "bursty"). However, most other 19th-century novels don't use the name at all.
The Chi-Square test, being sensitive to the large total count, flags it as significant. The Mann-Whitney test, which relies on ranks across many chunks, correctly identifies that its usage is not a general feature of 19th-century novels but rather an outlier concentrated in a small part of the corpus. Therefore, "Bartholomew" is distinctive of a few specific books, but not of the 19th-century corpus as a whole.